



# RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

**Department of Computer Science and Engineering**

**Academic Year: 2025-2026 (Even Semester)**

## INNOVATIVE TEACHING PRACTICES

**Degree, Semester & Branch:** IV Semester B.E. CSE

**Course Code & Title:** CS3452 & Theory of Computation

**Name of the Faculty member:** Dr. S. Poornam

**Date & Time:** 09.04.2026 & 09:35 A.M

**Topic:** Tractable and Intractable Problems

**Course Outcome:** CO5

**Type of Activity:** Mind Map

### **Justification**

The concepts of tractable and intractable problems in Theory of Computation involve understanding computational complexity, polynomial-time solvability, NP-completeness, and optimization challenges. Students often find it difficult to differentiate between various complexity classes and understand their interrelationships.

The Mind Map activity was introduced to visually organize concepts, improve conceptual understanding, and encourage collaborative learning. Mind mapping enabled students to connect related topics such as P, NP, NP-Complete, and NP-Hard problems systematically. The activity promoted active participation, analytical thinking, visualization skills, and better retention of theoretical concepts.

### **Activity Description**

#### **Pioneers to Peers Activity – Turing Machine**

##### **Introduction Phase**

- ❖ The instructor introduced the history and importance of Turing Machines in computation theory.
- ❖ Students were briefed about pioneering researchers and the evolution of computation models.
- ❖ The class was divided into small peer groups.
- ❖ Each group was assigned a specific Turing Machine concept or application.

##### **Peer Learning and Exploration Phase**

- ❖ Students explored:
  - Structure of Turing Machine,
  - Tape and head movement,
  - State transitions,
  - Acceptance and rejection conditions,
  - Computation process and problem-solving.

- ❖ Groups studied contributions of pioneers in automata and computation theory.
- ❖ Students prepared charts, state diagrams, and step-by-step tape simulations.
- ❖ Peer leaders guided discussions and explained concepts to teammates.

### **Presentation and Demonstration Phase**

- ❖ Each group presented:
  - Historical background of computation pioneers,
  - Working principle of Turing Machines,
  - Sample Turing Machine problems and tape operations.
- ❖ Students demonstrated:
  - Symbol replacement,
  - Left and right tape movement,
  - Transition function execution.
- ❖ Peer groups compared different computational approaches and applications.

### **Reflection and Assessment Phase**

- ❖ Students solved small Turing Machine construction problems individually.
- ❖ A reflective discussion was conducted on:
  - ✓ Challenges in understanding machine computation,
  - ✓ Importance of Turing Machines in modern computing,
  - ✓ Benefits of peer learning and collaborative exploration.

### **CO – PO / PSO Mapping**

| CO  | PO1 | PO2 | PO3 | PO4 | PSO1 |
|-----|-----|-----|-----|-----|------|
| CO1 | 2   | 2   | 2   | 1   | 1    |

*(1 – Low, 2 – Moderate, 3 – High)*

**PO/PSO mapped:** PO1, PO2, PO3, PO4, PSO1

### **Justification for correlation**

- ❖ Students applied theoretical concepts of Turing Machines to solve computational problems.
- ❖ Peer learning improved communication, collaboration, and presentation skills.
- ❖ Students developed analytical reasoning through machine simulation and state transition analysis.
- ❖ Group exploration enhanced confidence in understanding abstract computation concepts.
- ❖ The activity strengthened logical thinking and computational problem-solving abilities.
- ❖ Turing Machine concepts support advanced applications in AI, compiler design, algorithm analysis, and research-oriented computing.



Images / Screenshot of the practice:

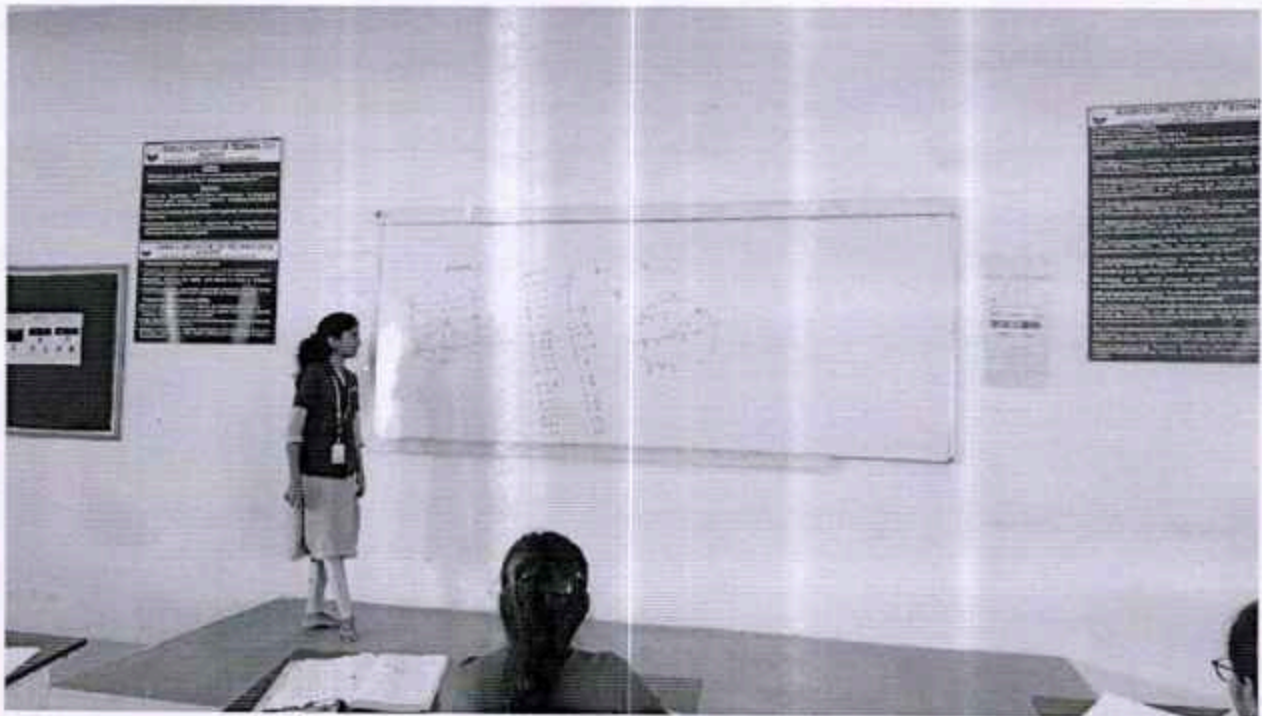


Fig:1 Tractable and Intractable Problems Concepts - SAMEEKSHA A (953624104133)

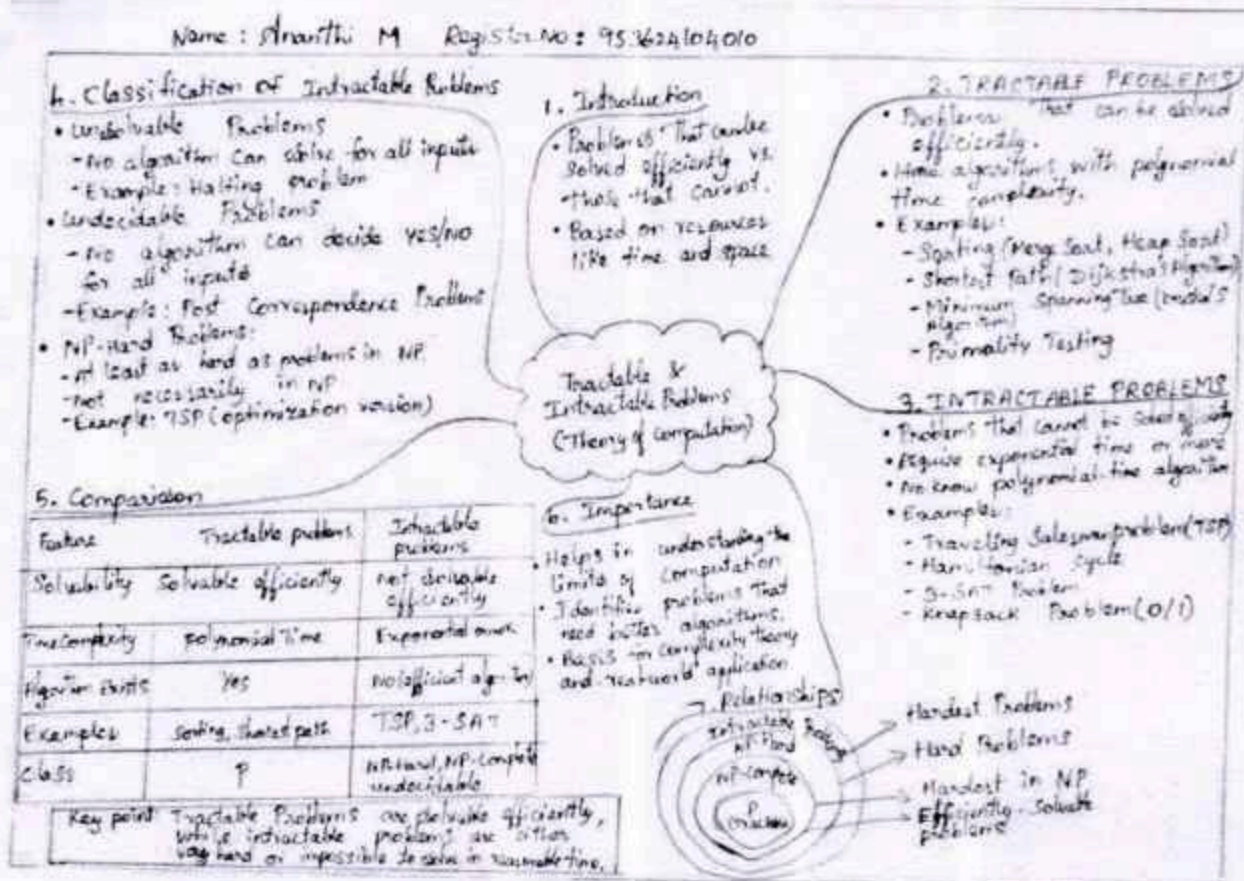


Fig 2: Hand drawn Mind Map Visualizations by - ANANTHI M (953624104010)

## Outcome

- ❖ Students demonstrated improved understanding of Turing Machine operations and computation logic.
- ❖ Peer-supported discussions enhanced conceptual clarity and active participation.
- ❖ Students gained confidence in solving automata-related computational problems.
- ❖ Collaborative learning improved teamwork and presentation skills.
- ❖ The activity promoted interactive and engaging learning of abstract computation concepts.

## Reflective Report

### Pre-Implementation

#### Identified Challenges

- Students found Turing Machine operations abstract and difficult to visualize.
- Difficulty in understanding tape movement and transition execution.
- Limited classroom interaction during traditional lecture sessions.

#### Addressing Challenges

- Introduced peer-led collaborative learning strategies.
- Encouraged student presentations and demonstrations.
- Used diagrams and tape simulations for better visualization.

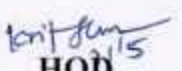
### Post-Implementation

- Students actively participated in peer discussions and demonstrations.
- Group activities improved analytical thinking and conceptual understanding.
- Most students were able to simulate simple Turing Machines independently.
- Peer interaction clarified misconceptions effectively.
- Feedback indicated that collaborative learning made Theory of Computation concepts more engaging and understandable.

## References

- Author and Citation Information for "Turing Machines"
- JFLAP Tutorial
- Implementation | Centre for Teaching Excellence | University of Waterloo

  
Signature of Faculty Member

  
HOD